SOP-BUF-001 Preparation of Tris-NaCl-EDTA (TNE) Buffer, pH 8.0

DEMO DATA - written for the SmartLab AI demonstration. This is not an approved laboratory procedure.

1. Purpose

This procedure describes the preparation of TNE buffer containing Tris base, sodium chloride and EDTA disodium salt dihydrate. The mass of each component is calculated from its molecular weight, the target concentration and the final volume required.

2. Safety

Wear safety glasses, a lab coat and nitrile gloves. Hydrochloric acid is corrosive: add it slowly, in a fume hood, and never add water to concentrated acid. Tris base and EDTA powders are irritants; avoid creating dust when weighing.

3. Procedure

Weigh the calculated mass of Tris base, sodium chloride and EDTA disodium salt dihydrate on a calibrated balance using a clean weighing boat. Transfer the solids to a clean beaker and dissolve them in deionised water equal to about 80% of the final volume, stirring with a magnetic stirrer. Calibrate the pH meter before use. Adjust the pH to 8.0 by adding hydrochloric acid dropwise while stirring. Transfer the solution to a volumetric flask and make up to the final volume with deionised water. Filter through a 0.22 um membrane filter into a sterile, labelled bottle.

4. Storage and Labelling

Label the bottle with the buffer name, pH, date and your initials. Store at 4 C and use within 1 month. Discard the buffer if it becomes cloudy.